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USE MATHEMATICS TO INVESTIGATE AND MONITOR THE FINANCIAL ASPECTS OF PERSONAL, BUSINESS,NATIONAL AND INTERNATIONAL ISSUES

US 7468

Summative assessment

Question 1

I. Simple vs Compound interest difference for 1 year

P = R1 000 000, i = 10%, n = 1 year

Simple Interest:I = P ×i ×n = 1000000 × 0.10 × 1 = R100 000

Compound Interest:A = P(1+i)^n = 1000000(1.10)^1 = R1 100 000 → Interest = R100 000

Difference:R100 000 - R100 000 = R0

For 1 year, simple and compound give the same interest. The difference only shows from year 2 onwards.

II. Annuity due - amount to invest each year

Nina wants R800 000 in 4 years. Payments at _beginning_ of each year = annuity due.

Formula: FV =A × [(1+i)^n - 1]/i ×(1+i)

We don't have i, but assuming 10% like above since no rate given. If different rate, tell me.

Using 10%:

800000 =A×[(1.10)^4 - 1]/0.10 ×1.10

800000 = A× [0.4641/0.10] × 1.10 =A ×5.1051

A= 800000 / 5.1051 = R156 670.039

Nina must invest ≈ R156 671.039 at the start of each year

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III. Present value of annuity due

A= R13 000, n=8, i=10%, paid at beginning

Formula: PV = A×[1 - (1+i)^{-n}]/i \times (1+i)

PV = 13000× [1 - (1.10)^{-8}]/0.10 ×1.10

PV = 13000 × [1 - 0.4665]/0.10 ×1.10 = 13000 ×5.3349 ×1.10

PV = 13000 ×5.8684 = R76 289.20

Prsent Value ≈ R76 289

Question 2

Financial instruments James can use:

1. Stocks/Shares - Buy ownership in companies for capital growth + dividends

2. Bonds - Government or corporate bonds for fixed interest income, lower risk

3. Unit Trusts/Mutual Funds - Pool money managed by professionals, diversified

4. Fixed Deposits- Bank deposit for guaranteed interest, low risk

5. Retirement Annuity / Tax-free savings account - Long term growth with tax benefits

Question 3

Loan = R400 000, i = 1% per month, n = 20 years = 240 months

I. Monthly repayment

Formula: A= P ×[i(1+i)^n]/[(1+i)^n - 1]

$PMT = 400000 ×[0.01(1.01)^{240}]/[(1.01)^{240} - 1]

(1.01)^{240} = 10.8926

A= 400000 × [0.108926]/[9.8926] = 400000 × 0.011011 = R4 404.40

II. Total interest over 20 years

Total paid = 4404.40 x 240 = R1 057 056

Interest = 1 057 056 - 400 000 = *R657 056*

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III. Monthly repayment for 15 years= 180 months

(1.01)^{180} = 5.9958

A = 400000 \times [0.01 × 5.9958]/[4.9958] = 400000 ×0.011996 = R4 798.40

IV. Total interest over 15 years

Total paid = 4798.40 x 180 = R863 712

Interest = 863 712 - 400 000 = R463 712

V. Interest saved

657 056 - 463 712 = R193 344 saved

Question 4

Marginal cost

$C(x)=10x+100x^{½}+10000$

$C'(x)=10(1)+100(½)x^{-½}+0$

$=10+50/x^{½}$

$C'(100)=10+50/100^{½}$

$=10+50/10$

$=10+5$

$C'(100)=15$

Marginal revenue

$P=1000/x^{½}$

$=1000x^{-½}$

$R(x)=p×x$

$=(1000/x^{½})× x$

$=1000× x^{1 - ½}$

$= 1000x^{½}$

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R'(x)=1000(½) x ⁻ ½

=500/x½

=500/ 100½

= 500/ 10

R'(x)=50

Question 6

Methods to control costs and optimise profits:

1. Budgeting & Variance analysis- Set cost targets and compare actual vs budgeted to find waste
2. Cost-volume-profit analysis- Find break-even point and optimal production level
3. Lean production / Waste reduction - Cut unnecessary processes and inventory
4. Negotiating with suppliers- Get better prices for inputs to lower MC
5. Revenue optimisation- Pricing strategies and product mix to maximize MR
6. Monitoring overheads- Control fixed costs and automate where possible